Lecture 28: Householder Reflectors; Reduction to Hessenberg Form

Math 6610

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Today's Goals

- Householder reflectors
- Reduction of A to Hessenberg form

Householder Reflectors

Definition (*Householder Reflectors*). $Q \in \mathbb{C}^{n\times n}$ of the form $Q = I - 2vv^H$, where $v \in \mathbb{C}^n$ with $\|v\|_2 = 1$, is a *Householder reflector* matrix.

Q is *Hermitian*:

$$Q^H = I - 2(vv^H)^H = Q.$$

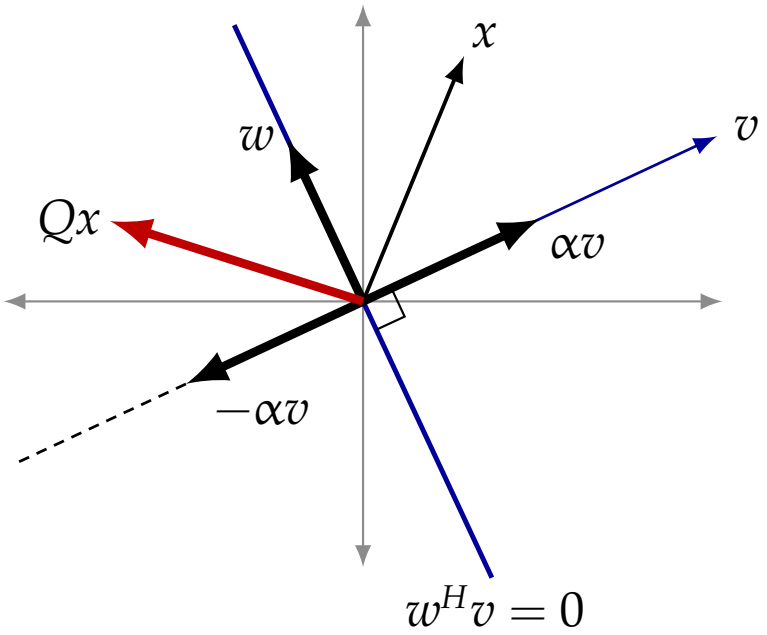
Q is unitary:

$$Q^H Q = Q^2 = I - 4vv^H + 4v \underbrace{v^H v}_{=\|v\|_2^2=1} v^H = I.$$

Householder Reflectors: Reflection

For any $x \in \mathbb{C}^n$:

$Qx =$ reflection of x w.r.t. the $(n-1)$-dimensional plane $\left\{w \in \mathbb{C}^n \;\middle|\; w^H v = 0\right\}$



Householder Reflectors: Reflection (Proof)

Proof.

$$
\begin{aligned}
x &= \alpha v + w, \qquad \text{where } \alpha \in \mathbb{C},\ w^H v = 0 \\
Qx &= (I - 2vv^H)(\alpha v + w) \\
&= \alpha v + w - 2v \underbrace{v^H \alpha v}_{= \alpha v^H v = \alpha} - 2v \underbrace{v^H w}_{=0} \\
&= -\alpha v + w
\end{aligned}
$$

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Constructing Q with $Qx = \gamma e_1$

Given: $x \in \mathbb{C}^n$.
We can construct a Householder reflector $Q = I - 2vv^H$ such that

$$\begin{aligned} Qx &= \gamma e_1 \qquad \text{for some } \gamma \in \mathbb{C} \text{ with } |\gamma| = \|x\|_2 \\ x &= \alpha v + w, \qquad w^H v = 0 \\ Qx &= -\alpha v + w = \gamma e_1 \\ x - \gamma e_1 &= \alpha v + w - \gamma e_1 = 2\alpha v \end{aligned}$$

Set

$$\tilde{v} := x - \gamma e_1 = \begin{bmatrix} x_1 - \gamma \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \qquad \text{and} \qquad v := \frac{\tilde{v}}{\|\tilde{v}\|_2}$$

Choice of γ; Summary

Numerically best choice of γ:

$$\gamma := -(\operatorname{sgn} x_1)\,\|x\|_2 \qquad \text{where} \qquad \operatorname{sgn} x_1 := \begin{cases} \dfrac{x_1}{|x_1|} & x_1 \neq 0 \\ 1 & x_1 = 0 \end{cases}.$$

Summary:

$$Q = I - 2vv^H, \qquad \text{where } v = \frac{\tilde{v}}{\|\tilde{v}\|_2}$$

$$\tilde{v} = x + (\operatorname{sgn} x_1)\,\|x\|_2\, e_1 = \begin{bmatrix} x_1 + (\operatorname{sgn} x_1)\,\|x\|_2 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$Qx = \gamma e_1, \qquad \text{where } \gamma = -(\operatorname{sgn} x_1)\,\|x\|_2$$

Hessenberg Matrix

Definition (*Hessenberg Matrix*).

http://en.wikipedia.org/wiki/Hessenberg_matrix

An *upper Hessenberg matrix* has zero entries below the first subdiagonal, and a *lower Hessenberg matrix* has zero entries above the first superdiagonal. For example:

$$\begin{bmatrix} 1 & 4 & 2 & 3 \\ 3 & 4 & 1 & 7 \\ 0 & 2 & 3 & 4 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$ is upper Hessenberg.

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 5 & 2 & 3 & 0 \\ 3 & 4 & 3 & 7 \\ 5 & 6 & 1 & 1 \end{bmatrix}$$ is lower Hessenberg.

Reduction of A to Hessenberg Form: Schur vs. H

Schur factorization:

$$U^H A U = T = \begin{bmatrix} * & \cdots & * \\ & \ddots & \vdots \\ 0 & & * \end{bmatrix}$$

Next "best" thing:

$$U^H A U = H = \begin{bmatrix} * & * & \cdots & \cdots & * \\ * & * & \ddots & & \vdots \\ & * & \ddots & \ddots & \vdots \\ & & \ddots & \ddots & * \\ 0 & & & * & * \end{bmatrix}$$

Hessenberg Form with $n-2$ Householder Reflections

Can be achieved with $n-2$ Householder reflections

$$U = U_1 U_2 \dots U_{n-2},$$

where $U_j = \begin{bmatrix} I_j & 0 \\ 0 & Q_{n-j} \end{bmatrix}$, $Q_{n-j} \in \mathbb{C}^{(n-j)\times(n-j)}$, Householder reflector

Example: $n = 5$

Example ($n = 5$).

$$A = \begin{bmatrix} a_{11} & * \\ x & * \end{bmatrix}$$

where $x \in \mathbb{C}^4$, $Q_1 \in \mathbb{C}^{4\times 4}$, $Q_1 x = \gamma_1 e_1$, $U_1 = \begin{bmatrix} 1 & 0 \\ 0 & Q_1 \end{bmatrix} = U_1^H$.

$$U_1^H A U_1 = U_1 A U_1 = (U_1 A) U_1 = \begin{bmatrix} a_{11} & * \\ \gamma_1 e_1 & * \end{bmatrix} U_1$$

Example: $n = 5$

Example ($n = 5$, *continued*).

$$U_1^H A U_1 = \begin{bmatrix} a_{11} & \# & \# & \# & \# \\ \gamma_1 & \# & \# & \# & \# \\ 0 & \# & \# & \# & \# \\ 0 & \# & \# & \# & \# \\ 0 & \# & \# & \# & \# \end{bmatrix}$$

$$U_3^H U_2^H U_1^H A U_1 U_2 U_3 = \begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ 0 & * & * & * & * \\ 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * \end{bmatrix}$$

Reduction to Hessenberg Form: Eigenvectors

$$U^H A U = H, \qquad U = U_1 U_2 \dots U_{n-2}, \quad A \in \mathbb{C}^{n\times n}$$

1. $\Lambda(A) = \Lambda(H)$. Eigenvectors transform as follows:

$$Ax = \lambda x,\ x \neq 0 \ \Leftrightarrow\ \underbrace{U^H A U}_{H} U^H x = \lambda U^H x$$

$$y = U^H x \ \Leftrightarrow\ Hy = \lambda y,\ y \neq 0$$

Here $y = U^H x$ and $x = Uy = U_1 U_2 \dots U_{n-2} y$,

$$U_j = \begin{bmatrix} I_j & 0 \\ 0 & Q_{n-j} \end{bmatrix}, \qquad Q_{n-j} = I - 2 v_{n-j} v_{n-j}^H, \quad v_{n-j} \in \mathbb{C}^{n-j}$$

Reduction to Hessenberg Form: Cost

2. Reduction of $A \in \mathbb{C}^{n \times n}$ to upper-Hessenberg form requires $\sim \frac{10}{3} n^3$ flops.

Left to do: Compute the eigenvalues of H.

The QR Algorithm: QR Factorization

Definition (*QR Factorization*). *QR factorization* of $M \in \mathbb{C}^{n\times n}$:

$$M = QR, \qquad \text{where } Q \in \mathbb{C}^{n\times n} \text{ is unitary and } R \text{ is upper-triangular}$$

This is done via Gram-Schmidt.

The QR Algorithm

Remark (*QR Algorithm (for eigenvalue computations)*).

Input: upper-Hessenberg matrix $H \in \mathbb{C}^{n \times n}$

- Set $A^{(0)} := H$
- For $k = 1, 2, \ldots$
  - Choose a suitable "shift" $\mu_k \in \mathbb{C}$
  - Compute a QR factorization:

$$A^{(k-1)} - \mu_k I = Q^{(k)} R^{(k)}$$

  - Set $A^{(k)} := R^{(k)} Q^{(k)} + \mu_k I$
- end (k)

The QR Algorithm: Lemma

Lemma.

1. $A^{(k)} = (Q^{(k)})^H A^{(k-1)} Q^{(k)}, \ k = 1, 2, \ldots$
2. $\Lambda(H) = \Lambda(A^{(k)}), \ k = 0, 1, 2, \ldots$

Proof.

2. By (1), $A^{(k)}, A^{(k-1)}, A^{(k-2)}, \ldots, A^{(1)}, A^{(0)} = H$ are all similar.

1.
$$A^{(k)} = R^{(k)} Q^{(k)} + \mu_k I = (Q^{(k)})^H \big(\underbrace{Q^{(k)} R^{(k)} + \mu_k I}_{= A^{(k-1)}} \big) Q^{(k)}$$

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Note: All the matrices $A^{(k)}$ are upper-Hessenberg!

Looking Ahead

- **Next lecture:** Givens Rotations